System Verification and Validation Plan for Software Engineering

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Revision History

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[The intention of the VnV plan is to increase confidence in the software. However, this does not mean listing every verification and validation technique that has ever been devised. The VnV plan should also be a **feasible** plan. Execution of the plan should be possible with the time and team available. If the full plan cannot be completed during the time available, it can either be modified to "fake it", or a better solution is to add a section describing what work has been completed and what work is still planned for the future. —SS]

[The VnV plan is typically started after the requirements stage, but before the design stage. This means that the sections related to unit testing cannot initially be completed. The sections will be filled in after the design stage is complete. the final version of the VnV plan should have all sections filled in. —SS]

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List of Tables

[Remove this section if it isn't needed —SS]

List of Figures

[Remove this section if it isn't needed —SS]

1 Symbols, Abbreviations, and Acronyms

symbol	description
T	Test

[symbols, abbreviations, or acronyms — you can simply reference the SRS (**?**) tables, if appropriate —SS]

[Remove this section if it isn't needed —SS]

This document ... [provide an introductory blurb and roadmap of the Verification and Validation plan —SS]

2 General Information

2.1 Summary

The Troitsky Toolbox project seeks to create a centralized information repository and a collection of software services that provide technical support for the McMaster Troitsky club. These services ensure information transfer between years and teams, and facilitate support and onboarding for new members. Served as a web application, accessible by all current club members, the services provided will simplify the technical components of the bridge design process, decreasing the barrier to becoming a competitive team.

One of the primary services in the toolbox is the model analysis service, which allows teams to load STEP files of their bridge designs and perform basic analysis on the models. This service will provide teams with a simplified analysis of their design, when compared to external static model analysis software, and will integrate with the material testing component and the rule compliance checker to provide a more specified analysis given the build material properties and current competition rules. Services regarding the information repository will include the ability to search through past designs and load them into the model analysis service, allowing teams to upload their designs with exit documentation and providing customized material testing result upload features. A knowledge transfer service collects executive handoff documents so that club knowledge carries over between years. In managing the toolbox access, role-based access control (RBAC) will be integrated to ensure that information is kept within the club, limiting access, and allowing executive members the ability to manage club users.

2.2 Objectives

The primary objective of the verification and validation plan is to outline the approach to ensure that the Troitsky Toolbox project will meet the requirements outlined in the software requirements specification. This plan will help to build confidence in the correctness, reliability and safety of the system, focusing verification on high-risk points and safety-critical concerns outlined

in the hazard analysis.

The V&V plan will primarily focus on these system objectives outlined below. There are also goals outlined that have been deemed out of scope for the current development cycle, but will be outlined in documentation as goals for future year development.

2.2.1 Primary Objectives

- **Adequate model analysis correctness:** Testing the correctness of the model analysis service is vital to the usability of the toolbox, as, without ensured correctness, teams will not be able to rely on the results. These results will not only be used by teams in the bridge design process, but will also guide them in making estimates of their bridge's weight and ultimate load capacity.

- **Data integrity and persistence:** The system must ensure data integrity and security in the knowledge repository. The repository will serve as the club's main data storage and information transfer, so losing data would be a critical concern and would defeat the project's purpose.

- **Access control:** The RBAC in the club will maintain data privacy both between teams and outside of the club. Testing this access control will provide confidence to teams when storing current bridge designs in the system, and ensure that executive and exit documentation is private to the club.

- **Demonstrating functional usability:** The functional usability of the system determines if it will be fully integrated into the club's organization and individual McMaster teams' bridge design and development process.

2.2.2 Out-of-Scope Objectives

- **Industry standard static model analysis performance:** The model analysis service is intentionally designed to provide a simplified approximation of static model analysis, so its accuracy will have a larger tolerance than industry-standard tools and will not aim to match established services.

- **Full year competition testing:** The toolbox testing will not include an end-to-end full-year competition evaluation within the current development cycle. This would require a usable product from the beginning of the school term, allowing teams to use the service throughout their full design process.

2.2.3 Assumptions

There is an assumption of the correctness of external libraries (BFE.NET, IxMilia.Step, occt-import-js, three.js, SQL Server) used. It is assumed that they are verified by their maintainers; only our integration with them is tested.

2.3 Custom Documents

2.3.1 Fall - Interviews

The primary CD for the fall term development is interviews conducted with core stakeholders: experienced Troitsky club members and executives. These semi-structured interviews will help outline expectations and prioritization of services from knowledgeable members. These interviews will be anonymized to ensure user privacy and aggregated into usable results for the toolbox development process. The interviews will be vital in checking any assumptions made in the SRS with experienced club members.

In addition to the core interviews for the fall CD, a survey has been performed to begin the requirements elicitation process. The survey design record and raw data are stored in the project repository, and the results were used in the SRS (**?**) to outline the priority of services and core issues members are currently facing.

2.3.2 Winter - Usability Report

The CD completed in the winter term will be a usability report, allowing for a more complete assessment of the functional usability of the toolbox. This round of testing will be performed with a more complete iteration of the product, allowing for more complete analysis of different system services' usability. Users will be gathered spanning different experience levels, both in club experience and technical experience (modelling), creating valuable data about the system's performance. Specific user functions will be tested, with

tasks that follow the G.5 usage scenarios, measured by completion rate and time on task.

This CD will further strengthen the V&V of the system, ensuring that complete functional and non-functional requirements come together into a fully usable service.

2.4 Relevant Documentation

1. **Software Requirements Specification (?):** The SRS defines the functional and non-functional requirements for the Troitsky Toolbox, and is referenced in the V&V plan as the standard the system wishes to meet. The requirements set the goals for the testing areas, identifying the importance of different services, which is then reflected in the testing performed for those given systems.

2. **Hazard Analysis (?):** The hazard analysis provides information regarding the primary risks and failure points of the system, pointing to the components that will require more in-depth and complex system testing. The hazard analysis also defines safety requirements for the system which are referenced in the V&V plan specifically with regard to system safety and access control correctness.

3 Plan

[Introduce this section. You can provide a roadmap of the sections to come. —SS]

3.1 Verification and Validation Team

[Your teammates. Maybe your supervisor. You should do more than list names. You should say what each person's role is for the project's verification. A table is a good way to summarize this information. —SS]

3.2 SRS Verification

[List any approaches you intend to use for SRS verification. This may include ad hoc feedback from reviewers, like your classmates (like your primary reviewer), or you may plan for something more rigorous/systematic. —SS]

[If you have a supervisor for the project, you shouldn't just say they will read over the SRS. You should explain your structured approach to the review. Will you have a meeting? What will you present? What questions will you ask? Will you give them instructions for a task-based inspection? Will you use your issue tracker? —SS]

[Maybe create an SRS checklist? —SS]

3.3 Design Verification

[Plans for design verification —SS]

[The review will include reviews by your classmates —SS]

[Create a checklists? —SS]

3.4 Verification and Validation Plan Verification

[The verification and validation plan is an artifact that should also be verified. Techniques for this include review and mutation testing. —SS]

[The review will include reviews by your classmates —SS]

[Create a checklists? —SS]

3.5 Implementation Verification

[You should at least point to the tests listed in this document and the unit testing plan. —SS]

[In this section you would also give any details of any plans for static verification of the implementation. Potential techniques include code walkthroughs, code inspection, static analyzers, etc. —SS]

[The final class presentation in CAS 741 could be used as a code walkthrough. There is also a possibility of using the final presentation (in CAS741) for a partial usability survey. —SS]

3.6 Automated Testing and Verification Tools

[What tools are you using for automated testing. Likely a unit testing framework and maybe a profiling tool, like ValGrind. Other possible tools include a static analyzer, make, continuous integration tools, test coverage tools, etc. Explain your plans for summarizing code coverage metrics. Linters are another important class of tools. For the programming language you select,

you should look at the available linters. There may also be tools that verify that coding standards have been respected, like flake9 for Python. —SS]

[If you have already done this in the development plan, you can point to that document. —SS]

[The details of this section will likely evolve as you get closer to the implementation. —SS]

3.7 Software Validation

[If there is any external data that can be used for validation, you should point to it here. If there are no plans for validation, you should state that here. —SS]

[You might want to use review sessions with the stakeholder to check that the requirements document captures the right requirements. Maybe task based inspection? —SS]

[For those capstone teams with an external supervisor, the Rev 0 demo should be used as an opportunity to validate the requirements. You should plan on demonstrating your project to your supervisor shortly after the scheduled Rev 0 demo. The feedback from your supervisor will be very useful for improving your project. —SS]

[For teams without an external supervisor, user testing can serve the same purpose as a Rev 0 demo for the supervisor. —SS]

[This section might reference back to the SRS verification section. —SS]

4 System Tests

[There should be text between all headings, even if it is just a roadmap of the contents of the subsections. —SS]

4.1 Tests for Functional Requirements

[Subsets of the tests may be in related, so this section is divided into different areas. If there are no identifiable subsets for the tests, this level of document structure can be removed. —SS]

[Include a blurb here to explain why the subsections below cover the requirements. References to the SRS would be good here. —SS]

4.1.1 Area of Testing1

[It would be nice to have a blurb here to explain why the subsections below cover the requirements. References to the SRS would be good here. If a section covers tests for input constraints, you should reference the data constraints table in the SRS. —SS]

Title for Test

1. test-id1

   Control: Manual versus Automatic

   Initial State:

   Input:

   Output: [The expected result for the given inputs. Output is not how you are going to return the results of the test. The output is the expected result. —SS]

   Test Case Derivation: [Justify the expected value given in the Output field —SS]

   How test will be performed:

2. test-id2

   Control: Manual versus Automatic

   Initial State:

   Input:

   Output: [The expected result for the given inputs —SS]

   Test Case Derivation: [Justify the expected value given in the Output field —SS]

   How test will be performed:

4.1.2 Area of Testing2

...

4.2 Tests for Nonfunctional Requirements

[The nonfunctional requirements for accuracy will likely just reference the appropriate functional tests from above. The test cases should mention reporting the relative error for these tests. Not all projects will necessarily have nonfunctional requirements related to accuracy. —SS]

[For some nonfunctional tests, you won't be setting a target threshold for passing the test, but rather describing the experiment you will do to measure the quality for different inputs. For instance, you could measure speed versus the problem size. The output of the test isn't pass/fail, but rather a summary table or graph. —SS]

[Tests related to usability could include conducting a usability test and survey. The survey will be in the Appendix. —SS]

[Static tests, review, inspections, and walkthroughs, will not follow the format for the tests given below. —SS]

[If you introduce static tests in your plan, you need to provide details. How will they be done? In cases like code (or document) walkthroughs, who will be involved? Be specific. —SS]

4.2.1 Area of Testing1

Title for Test

1. test-id1

   Type: Functional, Dynamic, Manual, Static etc.

   Initial State:

   Input/Condition:

   Output/Result:

   How test will be performed:

2. test-id2

   Type: Functional, Dynamic, Manual, Static etc.

   Initial State:

   Input:

Output:

How test will be performed:

4.2.2 Area of Testing2

...

4.3 Traceability Between Test Cases and Requirements

[Provide a table that shows which test cases are supporting which requirements. —SS]

5 Unit Test Description

[This section should not be filled in until after the MIS (detailed design document) has been completed. —SS]

[Reference your MIS (detailed design document) and explain your overall philosophy for test case selection. —SS]

[To save space and time, it may be an option to provide less detail in this section. For the unit tests you can potentially layout your testing strategy here. That is, you can explain how tests will be selected for each module. For instance, your test building approach could be test cases for each access program, including one test for normal behaviour and as many tests as needed for edge cases. Rather than create the details of the input and output here, you could point to the unit testing code. For this to work, you code needs to be well-documented, with meaningful names for all of the tests. —SS]

5.1 Unit Testing Scope

[What modules are outside of the scope. If there are modules that are developed by someone else, then you would say here if you aren't planning on verifying them. There may also be modules that are part of your software, but have a lower priority for verification than others. If this is the case, explain your rationale for the ranking of module importance. —SS]

5.2 Tests for Functional Requirements

[Most of the verification will be through automated unit testing. If appropriate specific modules can be verified by a non-testing based technique. That can also be documented in this section. —SS]

5.2.1 Module 1

[Include a blurb here to explain why the subsections below cover the module. References to the MIS would be good. You will want tests from a black box perspective and from a white box perspective. Explain to the reader how the tests were selected. —SS]

1. test-id1

   Type: [Functional, Dynamic, Manual, Automatic, Static etc. Most will be automatic —SS]

   Initial State:

   Input:

   Output: [The expected result for the given inputs —SS]

   Test Case Derivation: [Justify the expected value given in the Output field —SS]

   How test will be performed:

2. test-id2

   Type: [Functional, Dynamic, Manual, Automatic, Static etc. Most will be automatic —SS]

   Initial State:

   Input:

   Output: [The expected result for the given inputs —SS]

   Test Case Derivation: [Justify the expected value given in the Output field —SS]

   How test will be performed:

3. ...

5.2.2 Module 2

...

5.3 Tests for Nonfunctional Requirements

[If there is a module that needs to be independently assessed for performance, those test cases can go here. In some projects, planning for nonfunctional tests of units will not be that relevant. —SS]

[These tests may involve collecting performance data from previously mentioned functional tests. —SS]

5.3.1 Module ?

1. test-id1

   Type: [Functional, Dynamic, Manual, Automatic, Static etc. Most will be automatic —SS]

   Initial State:

   Input/Condition:

   Output/Result:

   How test will be performed:

2. test-id2

   Type: Functional, Dynamic, Manual, Static etc.

   Initial State:

   Input:

   Output:

   How test will be performed:

5.3.2 Module ?

...

5.4 Traceability Between Test Cases and Modules

[Provide evidence that all modules have been considered. —SS]

6 Appendix

This is where you can place additional information.

6.1 Symbolic Parameters

The definition of the test cases will call for SYMBOLIC_CONSTANTS. Their values are defined in this section for easy maintenance.

6.2 Usability Survey Questions?

[This is a section that would be appropriate for some projects. —SS]

7 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

7.1 AI Tools Used

[Give the name of the AI tool(s) used and the version. —SS]

7.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

7.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

8 Appendix — Reflection

[This section is not required for CAS 741 —SS]

The information in this section will be used to evaluate the team members on the graduate attribute of Lifelong Learning.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

2. What pain points did you experience during this deliverable, and how did you resolve them?

3. What knowledge and skills will the team collectively need to acquire to successfully complete the verification and validation of your project? Examples of possible knowledge and skills include dynamic testing knowledge, static testing knowledge, specific tool usage, Valgrind etc. You should look to identify at least one item for each team member.

4. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?